

# Depth — Master Reference for Claude Code

Sep 29, 2026 · @Phillip Argall

## How to use this document

This is the master brief for the next phase of Depth: it tells Claude Code what to build, in what order, and to what standard, for the hub, expeditions, Flats, the Deep Arcade with its four multiplayer games, and sound. Hand Claude Code one section at a time along with the existing design docs it extends.

### Documents this builds on

| Document                                                      | Status   | What this document does with it                                                                                                                          |
|---------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1_Depth_Overall.md                                            | Current  | Keeps its economy, controls, rendering ground rules, and design pillars; replaces the hub description and adds the arcade and the study hatch            |
| 2_Depth_Expeditions.md                                        | Current  | Keeps classes, abilities, relics, enemies, and bosses; adds the sonar chart, an AI that scales by tier, a visual and animation overhaul, and new systems |
| 3_Depth_Flats.md                                              | Current  | Keeps the rules, sigils, and catalogue; completes visuals, adds card physics, deeper systems, and a longer generated map                                 |
| 4_Depth_Parkour.md                                            | Finished | Untouched. Only its sound rules are reused                                                                                                               |
| Fathoms RTS design document ( Fathoms — RTS Design Document ) | Current  | The full design of arcade game 4, summarized in this document                                                                                            |

### Ground rules that apply to every section

- All art stays procedural (no image or audio files), inside the existing three visual languages: Darkest Dungeon for salon and expeditions, pixel art for parkour, Inscryption for Flats.
- Keep the existing engine rules from `1_Depth_Overall.md` section 5: `DrawTri()`, `RL_QUADS`, `BeginLayer / EndLayer`, supersampling with `SS = 2`.
- Every new number goes in `data.cpp` or a data file, never inline in logic.
- Every new system ships with a headless self-test option on `depth.exe`, like the existing `--sim`, `--flats-sim`, and `--verify` tools.
- Each section ends with a Definition of Done. Claude Code should treat it as the acceptance test before moving on.

**Reference images** (Darkest Dungeon): the Hamlet screen shows what the hub should feel like: a full scene with the interactable buildings placed in depth, the roster and resources framed at the edges, a single Embark button. The route map shows the branching room-and-corridor structure the sonar chart replaces. The combat screenshots show the standard for figures: chunky ink-outlined characters with heavy silhouettes, dramatic lighting, and a skill bar and stat panel below the scene.

## Shared visual direction: Darkest Dungeon, in code

The goal for the salon and expeditions is that a screenshot could pass for Darkest Dungeon at a glance, while every object is brass, water, and drowned myth. The current figure shader and `InkPass()` are the right foundation; what's missing is silhouette, lighting drama, layered depth, and secondary motion. This section defines the shared upgrades that the hub, expeditions, and enemies all use.

### What makes Darkest Dungeon read the way it does

| Trait                              | How to reproduce it in Depth                                                                                                                                 |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thick, variable-width ink outlines | Line weight in the figure shader scales with distance from the camera plane (thicker in front), doubles on the shadow side of a form, and thins on lit edges |

| Trait                                    | How to reproduce it in Depth                                                                                                                                                                                                          |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy silhouettes                        | Every character must read from its black silhouette alone: big hands and weapons, hunched shoulders, long torsos, small heads, wide stances. Add a <code>--silhouette</code> shot mode that renders figures solid black to check this |
| Hatching that follows form               | Cross-hatch direction follows each primitive's surface tangent, not the screen; hatch density is driven by the lightmap                                                                                                               |
| Painterly light                          | One strong key light per scene (the flashlight, a window, the candle), a cool fill from the water, and a rim light on the side away from the key. Store this as a <code>SceneLight</code> struct with key, fill, rim, and a fog color |
| Muted paper palette with hot accents     | Base tones are desaturated umbers, slates, and sea-greens; blood, glow, gold, and eyes get saturated color. Enforce with a per-location <code>Palette</code> (5 base tones, 2 accents) in <code>data.cpp</code>                       |
| Layered parallax with fog between layers | Between every two parallax layers, a fog quad tinted with the scene fog color, opacity growing with depth, so far layers fade like a stage set                                                                                        |
| Grain, vignette, and ink spatter         | <code>InkPass()</code> already adds grain and edge ink; add a vignette, a few scattered ink flecks fixed per <code>visSeed</code> , and a faint paper texture that never scrolls                                                      |
| Things in the frame move                 | Nothing static: candle flames, drips, dust in light beams, cloth in current, breathing figures, animals                                                                                                                               |

## Figure upgrades ( `render.cpp` , `enemyart.cpp` )

- **A shared rig.** Introduce `Rig`: named bones (root, hips, spine, chest, head, shoulders, elbows, wrists, knees, ankles, plus tails, tentacles, and props). Heroes and enemies are built by attaching the existing `ShadeLimb` / `ShadeBall` / `ShadeQuad` primitives to bones. This replaces per-class hard-coded limb positions with poses that can be blended and reused.

- **Clips.** `Clip` = keyframed bone curves with easing (ease-in anticipation, snap, overshoot, settle). Each figure runs a base clip (idle, walk, guard), an additive breathing clip, and reaction clips (hit, dodge, stagger, death) layered on top. Keep the existing `SpringPose` as the blend between clips so transitions overshoot naturally.
- **Secondary motion.** `Chain`: a short Verlet rope attached to a bone for hair, coat hems, harpoon lines, tentacles, chains, and dangling gear. Chains react to the parent's movement, to a per-scene current vector, and to hits. This is the single biggest step toward "alive".
- **Depth ordering.** Parts sort by bone depth so an arm crosses in front of the torso when it swings. Weapons and props are bones too.
- **Facial life.** Two-state eyes (open, shut) with a blink every 3–6 s, an eye direction that tracks the current target, and a mouth state used on crits, pain, and shouts.
- **Materials.** Tag each primitive as skin, cloth, metal, shell, glow, or wet. The shader varies specular (metal and wet shine, cloth doesn't), rim strength, and hatching density by material.
- **Vibrance outside combat** ( `uVibrance` ) stays: the salon is warmer than the deep.

## Backgrounds

- Keep the seven parallax layers, but each layer becomes a list of `Prop` objects (a rock, a pillar, a wreck) with their own idle motion (sway, drip, flicker) rather than a single painted band.
- Foreground occluders: at least one layer in front of the action (a rope, a stalactite, kelp, a railing), blurred and dark, to put the fighters *inside* the scene.
- Lightmap first, ink second, bright particles last (the existing rule). Volumetric beams: cones from the key light with dust particles drifting through them.

## Animation principles for every scene

1. Anticipation before action (a windup of 6–10 frames), a fast strike (2–3 frames), then overshoot and settle.
2. Hits push the target's root bone away and shake its chains; the camera nudges 2–4 px on heavy hits.
3. Nothing holds a pose for more than 2 s without a breath, a weight shift, or a glance.
4. Reactions are proportional: a graze is a flinch, a crit is a stagger with a spray of ink flecks.

## Definition of Done

- `--shots` renders every hero, enemy, and boss in idle, attack, hit, and death; a `--silhouette` pass shows each is identifiable in black.
- Every figure has at least one Chain and a blink.
- The three-light rig and per-location palette are applied in every location and the salon.
- A side-by-side of the salon and a Darkest Dungeon screenshot reads as the same style family.

## The hub: the Nautilus's grand salon, reimagined

The salon becomes Depth's Hamlet: one fixed, deep scene where every station is a real object placed in the room, framed by the roster on the right, resources at the bottom left, and one Embark button. It stays a single non-scrolling screen, but it stops being a flat cutout: the camera looks down the length of the salon toward the great window, with foreground objects cutting across the frame and the far wall receding into blue light.

## Composition

- **Camera.** Fixed at the salon's aft door, slightly above eye height, looking forward. A narrow field of view keeps the room claustrophobic: the riveted ceiling ribs press down, pipes and gauges run along the walls, condensation beads on brass.
- **Three depth bands.** Foreground (the card table's edge, a brass railing, hanging lamps), midground (most stations, the crew walking), far (the great window, the helm, the organ against the far wall). Fog between bands, per the shared visual rules.
- **Mouse parallax.** Layers shift up to 1.5% of screen width opposite the cursor, so the room has depth without a moving camera.
- **Light.** Warm oil lamps as the key light, the sea's blue as the fill through the great window, candle rim light at the card table. Caustics ripple on the ceiling from the window.

## Station placement

| Station           | Where in the room                                           | The object                                | Hover                                         | Click                                |
|-------------------|-------------------------------------------------------------|-------------------------------------------|-----------------------------------------------|--------------------------------------|
| The Helm (Embark) | Far center, in front of the great window                    | The wheel and chart table, lit by the sea | Wheel glints, a brass plaque reads "The Helm" | Location and tier panel, then Embark |
| Crew Quarters     | Midground left, a door with a brass "CREW" sign             | Half-open door, bunks visible             | Door swings wider                             | Party, loadouts, relics              |
| Radar Room        | Midground right, a sonar console with a sweeping scope      | Green sweep, blips                        | Sweep speeds up                               | Recruits and salvager relics         |
| The Periscope     | Far right, a periscope column with handles                  | Periscope turns slightly                  | Handles fold out                              | Parkour dive select (unchanged)      |
| Workshop          | Midground left, a bench with tools and a half-built machine | Sparks, a spinning gear                   | Gear spins faster                             | Ship upgrades                        |
| The Ward          | Midground right, a door with a red cross and a cot          | Cot, hanging bottles                      | Lamp brightens                                | Healing                              |
| Sick Bay          | Far left, the pipe organ with a chair beside it             | Organ, a rattled hand resting             | Organ plays two notes                         | Rest rattled crew                    |
| Library           | Left wall, floor-to-ceiling bookshelves with a ladder       | Books, a globe                            | Ladder slides                                 | Lore                                 |

| Station         | Where in the room                                                                 | The object                                              | Hover                               | Click                                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| The Card Table  | Foreground right, a round table with a candle and the dealer's chair              | The dealer shuffling; cards flutter                     | Candle flares                       | Flats                                                                                                                     |
| The Deep Arcade | Foreground left, the cabinet riveted into the bulkhead                            | Jellyfish marquee, porthole screen playing attract mode | Jellyfish brighten, gauges twitch   | Arcade menu (see the arcade section)                                                                                      |
| The Study hatch | Bottom center, a brass floor hatch with a ladder and a glowing down-arrow chevron | Hatch with a wheel lock                                 | Wheel turns, light rises from below | Descends to the Study (placeholder screen until built: a locked door with a brass "Under Refit" sign and a return ladder) |

Every station keeps a reason to be where it is: the helm faces the window, the workshop is by the engine bulkhead, the ward is by the sick bay, the arcade is where the crew relax near the card table.

### Framing UI (Darkest Dungeon's edges)

- **Roster** on the right edge: portrait cards for every crew member with level pips, HP, nerves, and a status glyph (injured, rattled, ready). Drag a portrait onto the Helm's party slots. Rattled or injured crew are dimmed with an ink cross.
- **Resources** at the bottom left: gold, ship batteries, stored relics, with the same ink-plate style as the combat HUD.
- **Embark** at the bottom center, red and brass, only lit when the party is full.
- **Panels** slide in from the station's side of the screen over a darkened room, as in Darkest Dungeon, never a separate scene.

## Salon life

- The two working hands and the cat stay (the user likes them). Rebuild the hands on the shared `Rig` with clips for their posts (polishing, charting, tea) and give each a Chain (apron, hair, a tool on a lanyard).
- Add a third role that only appears sometimes: a hand playing the arcade cabinet, glancing back when the player approaches.
- The dealer sits at the card table at all times, shuffling and occasionally flicking a card into the air.
- The cat may sit on the arcade cabinet or the chart table.
- Ambient motion: lamps sway with the ship's roll, the organ's pipes hum with a faint glow when the sick bay is in use, fish pass the great window, bubbles rise from the periscope well.

## Technical plan ( `salon.cpp` )

1. Replace the flat `PaintWall` art with `Prop` objects placed in 3D and projected with the existing `Proj()` ; walls remain painted, but everything interactable becomes a placed object.
2. Each station is a `Prop` with a screen-space hitbox computed after projection each frame; hover state drives its idle clip and the plaque.
3. Panels are overlays inside the salon scene, not new scenes, so the room stays visible behind them.
4. `--shots` gets a salon page for every hover state and every open panel.

## Definition of Done

- The salon renders in three depth bands with fog and the three-light rig; the silhouette of the room reads as a submarine interior.
- All eleven stations are placed objects with hover motion, a plaque, and a panel.
- The arcade cabinet and the study hatch exist; the hatch leads to the placeholder Study and back.
- The roster edge, resources plate, and Embark button work with drag-and-drop party building.
- Hands, dealer, and cat run on rigs with chains and idle clips.

# Expeditions: the sonar chart

The linear corridor of rooms is replaced by a generated map of rooms and corridors, presented as the Nautilus’s sonar screen. Every expedition becomes a set of route decisions: the short path to the boss runs through fights; the loops carry treasure, curios, and rest, but burn light and nerves.

## Presentation

- The chart is a round phosphor sonar scope in the ship’s brass bezel, overlaying the walking scene when opened (Tab or the scope icon) and shown small in a corner while walking.
- Rooms are ink glyphs on the scope; corridors are faint dotted echoes. The party is a bright blip. Unvisited rooms show as a soft “?” echo until scouted or entered. A sweep passes every 2 s and briefly brightens what it crosses.
- The scope’s brightness follows the flashlight level: at low light the phosphor dims and noise crackles across it, so the map itself tells you the light is failing.
- Clicking an adjacent room sends the party walking; the walking scene plays the corridor with its events, then the room.

## Structure

| Tier (cave level) | Rooms | Corridor segments per corridor | Fights on the shortest path | Loops | Curios | Rest rooms |
|-------------------|-------|--------------------------------|-----------------------------|-------|--------|------------|
| 0                 | 6     | 2                              | 2                           | 1     | 2      | 1          |
| 1                 | 8     | 2-3                            | 3                           | 1     | 3      | 1          |
| 3                 | 10    | 3                              | 3                           | 2     | 4      | 1          |
| 5                 | 12    | 3-4                            | 4                           | 2     | 5      | 2          |
| 6                 | 14    | 4                              | 4                           | 3     | 6      | 2          |

**Room types** (extends `RoomType`): Entrance, Fight, Treasure (a chest, sometimes locked), Curio (an object to inspect, see New content), Rest (camp: heal, calm nerves, but a night ambush chance), Shrine (a location-specific event), Boss (always the farthest room from the entrance).

**Corridor segments** carry small events: nothing, a hallway fight (weaker group), a trap (bleed, poison, or nerves; a Diver or Whaler in the party spots traps 50% of the time and disarms them), a loose relic or loot, a curio, or a blocked passage that costs a battery to

clear. Each segment drains light as a room did before (20 / 16 / 12 / 9 by Reflector level, halved for revisited corridors).

### Generation rules ( `levelgen` for expeditions, seeded by the run)

1. Lay rooms on a loose grid, entrance on one edge, boss at maximum graph distance.
2. Build a spanning tree, then add the tier's loops so at least two distinct routes reach the boss.
3. Place the shortest route's fights, then push Treasure and Curio rooms off it by one or two rooms so loot always costs a detour.
4. Put Rest rooms roughly halfway to the boss and never adjacent to it.
5. Reject and reroll any map where the boss is reachable in fewer fights than the tier's minimum, or where more than two dead ends exist.

### Scouting and information

- Entering a room reveals rooms one step away as glyphs. The Sonar Array upgrade reveals two steps (level 2) or the whole layout without contents (level 3).
- Curios and the Awakened Lantern relic can reveal room contents. A Diver's *Mark the Prey* used out of combat scouts the next room's enemies.
- Backtracking is allowed; a revisited corridor has no scripted fight but a 15% ambush chance.

### Objectives

The Helm offers one objective per expedition, in Darkest Dungeon's manner, with a bonus on completion:

| Objective        | Requirement                | Bonus                   |
|------------------|----------------------------|-------------------------|
| Slay the boss    | Kill the boss              | +25% XP                 |
| Chart the depths | Visit 90% of rooms         | +50% gold and one relic |
| Salvage          | Open all Treasure rooms    | +1 relic                |
| Cleanse          | Win every fight on the map | +40% XP                 |

Retreat is allowed from any room; the party keeps loot but not the objective bonus, and each hero gains nerves.

## Definition of Done

- `depth.exe --gen-chart <tier> <seed>` prints a generated chart as ASCII and asserts the generation rules.
- The scope renders and updates during walking, with the light-driven noise.
- `--sim` runs full chart expeditions with route choice by the sensible auto-player; win rates are within 10 points of the current corridor numbers at each tier before tuning.

## Expeditions: visual overhaul and making them alive

Every fighter, enemy, boss, and background gets rebuilt on the shared rig and lighting rules so the combat screen reads like Darkest Dungeon: four heavy silhouettes facing four, a lit stage with fog behind and occluders in front, and a HUD of ink plates below.

### The combat stage

- **Layout** follows the reference screenshots: the scene fills the top two thirds; below it, the selected hero's portrait, stats, and skill bar on the left, the targeted enemy's panel with resistances on the right, and the round counter in a brass medallion at the top center.
- **Turn order** is a strip of small portraits above the HUD; the active fighter steps forward half a rank and gets a lit ground ring.
- **Camera.** Subtle push-in (3%) on the attacker during a windup, a snap back on impact, 2-4 px shake scaled to damage, and a slow drift during idle so the frame never freezes.
- **Light.** The flashlight stays the key light; at low light the fill drops, the rim rises, and the fog closes in, so danger is visible.

### Crew: look and silhouette

| Class   | Silhouette rule                                        | Signature motion                                    |
|---------|--------------------------------------------------------|-----------------------------------------------------|
| Nurse   | Tall and narrow, apron and satchel, a scalpel held low | Quick precise stabs, a step back after every action |
| Diver   | Round brass helmet, coiled harpoon line (a Chain)      | Low lunges, the line whips behind                   |
| Captain | Wide shoulders, long coat (Chain hem), cutlass raised  | Big overhead cuts, points when rallying             |

| Class        | Silhouette rule                                           | Signature motion                                  |
|--------------|-----------------------------------------------------------|---------------------------------------------------|
| Mechanic     | Squat and broad, a wrench as long as the torso            | Slow heavy swings with a full-body follow-through |
| Whaler       | Long harpoon gun, a slicker hood, rope coils              | Braces, fires, is kicked back by recoil           |
| Stowaway     | Hunched, bottle in hand, ragged sleeves                   | Stumbles, wild swings, a hiccup in idle           |
| Merman       | Sinuous, fins and a trident, tail Chain                   | Coiling then a whip-strike                        |
| Island Queen | Regal posture, a tall scepter, a feathered mantle (Chain) | Slow gestures, curses drawn in the air            |
| Robot        | Blocky, boiler chest, pistons; steam puffs                | Ratcheting motion, a stomp that shakes the stage  |
| Octopus      | Eight arm Chains, a bulbous head                          | Constant drifting arms, flurries that blur        |
| Siren        | Flowing hair and fins (Chains), hands raised to sing      | Sways, notes drawn as ink ripples                 |
| Wisp         | A floating lantern-spirit with trailing light             | Bobs, flares, and darts                           |

Each class gets the full clip set: idle, stressed idle, Death's Door idle, walk, guard, one windup and strike per attack family (slash, thrust, swing, shoot, throw, cast, heal, song), hit, dodge, stagger, crit reaction, death, and victory.

## Enemies and bosses

- Every enemy in `enemyart.cpp` moves to the rig with the same clip set, plus one signature idle each (the Sea Louse scuttles in place, the Siren's hair drifts, the Lost One Cultist's chains sway).
- **Bosses are staged like sets.** A level boss's ranks become independent parts on one rig: Cthulhu's wings beat, the tentacled face writhes, the sigil arm glows in rhythm; the Crustacean Queen's brood crawl on her carapace; Neptune's water spirals around the trident; the Sun God's mask cracks and glows as he takes damage. Bosses have a phase-change clip at 50% HP that changes their idle and palette.

- Mini-bosses get a two-rank rig with a distinct entrance clip (the Lobster surfaces from the silt; the Great White slides in from the fog).
- Death clips are unique per enemy family: crustaceans crack and spill, drowned things dissolve into bubbles, the tribe drops to a knee, Atlanteans crumble like statues.

## Backgrounds per location

| Location   | Far                                 | Mid props                                | Foreground occluders            | Living details                                                   |
|------------|-------------------------------------|------------------------------------------|---------------------------------|------------------------------------------------------------------|
| The Cave   | Ridges, water, glowing mould        | Coral heads, a wrecked hull, stalagmites | Stalactites, a hanging net      | Drips, mould pulse, crabs crossing                               |
| The Island | Eclipse sunset, palm isles          | Bone totems, huts, smoke                 | Palm fronds, spears in the sand | Fireflies, drums shaking a totem, a dog's eyes in the dark       |
| The Weeds  | Kelp forest to the vanishing point  | Kelp trunks, a sunken boat               | Kelp fronds swaying (Chains)    | Fish schools, bubbles, a shadow passing                          |
| Atlantis   | The drowned city under the pale eye | Columns, statues, cult banners           | A broken arch, chains           | The eye blinks, banners drift, sigils glow in time with the boss |

## Effects language

- Hits: an ink splash at the impact point, damage numbers that drop like weights, the target's Chains whip.
- Crits: a white flash frame, a bigger splash, the camera shake doubles, the stage dims for 0.2 s.
- Heals: warm glow rising, ink flecks turning gold. Nerves: a cold blue ripple and a brief hatch pattern over the portrait.
- Status effects: bleed drips from the figure, poison beads green, stun spins small brass gears, marks hang an ink crosshair.
- Death's Door: the figure's palette drops to grey with red rim light, breathing slows.

## Definition of Done

- Every class and enemy has the full clip set on the rig; `--shots` includes a sheet per figure showing all clips.
- Every boss has independently moving parts and a phase-change at 50% HP.
- Each location's background is built from props with idle motion and at least one foreground occluder.
- The combat HUD matches the layout above and no figure is ever motionless for more than 2 s.

## Expeditions: enemy AI that scales from tier 0 to tier 6

Every enemy chooses its action with one scoring brain, `EnemyBrain`, whose weights and lookahead grow with the tier. At cave level 0 it is nearly random; at level 6 it focuses fire, sets up combos, reads turn order, and punishes the mistakes a player should have learned to avoid by then.

### How a decision is made

1. List every legal (ability, target) pair for the acting enemy.
2. Score each pair as a weighted sum of the features below.
3. Pick with a softmax whose temperature falls with tier (random-ish at 0, near-greedy at 6). Ties and noise keep even level 6 from being fully predictable.

| Feature         | What it measures                                                      | Tier 0 | Tier 1 | Tier 3 | Tier 5 | Tier 6 |
|-----------------|-----------------------------------------------------------------------|--------|--------|--------|--------|--------|
| Expected damage | Damage × hit chance, minus the target's protection                    | 1.0    | 1.0    | 1.0    | 1.0    | 1.0    |
| Kill potential  | Bonus if the hit can drop the target to 0 or kill one at Death's Door | 0      | 0.5    | 1.5    | 2.5    | 3.0    |
| Focus fire      | Bonus for the ally-marked or already-wounded target                   | 0      | 0.6    | 1.2    | 1.5    | 1.5    |
| Threat          | Bonus against the hero with the highest damage dealt this fight       | 0      | 0      | 0.8    | 1.2    | 1.5    |
| Healer denial   | Bonus for stunning, pulling, or killing the party's healer            | 0      | 0      | 1.0    | 1.5    | 2.0    |

| Feature             | What it measures                                                                                            | Tier 0 | Tier 1 | Tier 3 | Tier 5  | Tier 6                     |
|---------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|--------|---------|----------------------------|
| Rank disruption     | Value of pulling a back-rank hero forward or shoving a melee hero back, weighed by which skills it disables | 0      | 0      | 0.5    | 1.5     | 2.0                        |
| Nerve pressure      | Stress damage weighted by how close the target is to Rattled                                                | 0.2    | 0.4    | 0.8    | 1.2     | 1.5                        |
| Status fit          | Penalty for bleeding a high-bleed-resist hero, poisoning a Robot, stunning a stunned target                 | 0      | 0.3    | 1.0    | 1.5     | 2.0                        |
| Setup               | Bonus for actions that enable an ally's better action next turn (pull, then bleed; mark, then heavy hit)    | 0      | 0      | 0.5    | 1.5     | 2.0                        |
| Self-preservation   | Heal or buff when an ally or self is below 40% HP; retreat to a back rank if cowardly                       | 0.2    | 0.5    | 1.0    | 1.2     | 1.2                        |
| Turn-order reading  | Prefer stunning the hero who acts next; avoid buffing when the buffed ally is about to die                  | 0      | 0      | 0      | 1.0     | 2.0                        |
| Lookahead           | Simulate one round with the sensible auto-player as the hero side and score the resulting state             | none   | none   | none   | 1 round | 1 round, best of 3 samples |
| Softmax temperature | Randomness                                                                                                  | 2.0    | 1.2    | 0.7    | 0.4     | 0.25                       |

## Personalities (per enemy family)

| Personality | Families                                                      | Behavior                                                   |
|-------------|---------------------------------------------------------------|------------------------------------------------------------|
| Swarm       | Sea Louse, War Dog, Fry                                       | Ignores self-preservation; all-in on the wounded           |
| Cunning     | Pistol Shrimp, Siren, Shaman, Cultist                         | Doubles healer denial and nerve pressure; hangs back       |
| Brute       | Dysformed Crustacean, Feral Merman, Lost One Infantry, Merman | Doubles expected damage and kill potential                 |
| Cowardly    | Brine Worm, Bilge creatures                                   | Retreats when below 50% HP; attacks when a hero is stunned |

| Personality | Families                      | Behavior                                                          |
|-------------|-------------------------------|-------------------------------------------------------------------|
| Guardian    | Armored Lost One, Coral Golem | Protects the boss or the highest-value ally, uses guard and taunt |
| Boss        | All bosses and mini-bosses    | Uses phase scripts (below) on top of the scoring                  |

## Boss scripts

Bosses combine the brain with a short script that changes at 50% HP:

- **Crustacean Queen:** phase 1 alternates Tidal Crush and Spawning Surge; phase 2 keeps two shrimp alive at all times and roars whenever three heroes are below half HP.
- **Sun God:** phase 1 burns the front; phase 2 opens every round with Aegis of Gold if any debuff is on him, then Wrath on the fastest hero.
- **Neptune:** pulls the healer forward, then targets that hero with Trident Impale next round.
- **Cthulhu:** counts madness stacks; when two heroes are mad, Gaze; otherwise Cosmic Crush on the marked or lowest-HP hero. Siphon Reality only when below 30%.

## What the player must have learned by tier 6

- Keep the healer out of pull range or guard them; enemies will go for them.
- Kill Cunning enemies first; they set up the Brutes.
- Watch the turn strip: level 6 enemies stun whoever acts next.
- Manage nerves before 80; late enemies push whoever is closest to Rattled.
- Match resistances: late enemies won't waste a bleed on the Robot, and neither should you.

## Definition of Done

- `depth.exe --brain-test <tier> <runs>` reports hero win rate against the same encounters at every tier; level 0 must be within 5 points of today's random AI and level 6 must lower the sensible auto-player's win rate by 15–25 points before rebalancing.
- Boss win-rate targets from `2_Depth_Expeditions.md` (66–85%, Cthulhu about 20%) are re-tuned after the brain lands, and the numbers are recorded in the balance changelog.
- Every enemy family has a personality entry in `data.cpp`.

# Expeditions: new content and systems

The roguelike grows from "walk, fight, boss" into a full Darkest Dungeon-style campaign layer: supplies, curios, camping, habits, ailments, resolve, crew bonds, skill training, a memorial, and two deeper tiers. Each system below is small on its own and reuses the existing `Status` , relic, and Ward code where it can.

| System       | What it adds                                                                                                                                                                                                                                                                                                                                                                       | Where it lives                                                   |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Provisioning | Before Embark, the Quartermaster sells supplies for the pack: batteries (25 gold), bandages (20), antivenom (20), grog rations (15, calms nerves at a rest room), a crowbar (30, opens locked chests and blocked passages without a key), salt (10, cleanses cursed curios). The Helm suggests a kit per location. Unused supplies are lost on return, so buying is a real choice  | Helm panel, <code>data.cpp</code>                                |
| Curios       | Inspectable objects in Curio rooms and corridors with weighted outcomes (loot, a relic, a buff, nerves, an ailment, an ambush). Using the right supply forces the good outcome. About 5 per location plus 6 shared. Examples: a barnacled sea chest (crowbar), a drowned sailor's locker (salt), a glowing anemone bed (bandage), a tribal idol (grog), a cult altar (no safe use) | New <code>Curio</code> table; outcomes reuse <code>Status</code> |

| System                            | What it adds                                                                                                                                                                                                                                                                                                                                                            | Where it lives                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Rest rooms and camp skills</b> | At a Rest room the party camps: 4 supply points of camp skills (each class has 2: the Nurse's Triage Tent heals 3 HP to all; the Captain's Sea Shanty removes 15 nerves from all; the Mechanic's Barricade halves the night-ambush chance; the Siren's Lullaby prevents nerve gain next fight, and so on). A 20% night-ambush fight can interrupt                       | New <code>CampSkill</code> per class                           |
| <b>Habits (quirks)</b>            | Each hero has up to 3 positive and 3 negative habits, gained from expeditions and curios: Steady Hands (+5 accuracy), Deck Legs (+2 dodge on the Weeds), Night Eyes (less nerve gain in low light), Bottle Fiend (drinks any grog found), Claustrophobe (+nerves in the Cave), Superstitious (must inspect idols). Habits can be locked or removed at the Ward for gold | <code>Hero::habits</code>                                      |
| <b>Ailments</b>                   | Diseases from curios and some enemies: Salt Rot (−3 max HP), Reef Fever (−2 speed), Barnacle Lung (+20% nerves), the Bends (−5 accuracy). Cured at the Ward for 60–120 gold or by a Nurse's Bone Saw out of combat                                                                                                                                                      | Reuses <code>Status</code> with a <code>persistent</code> flag |
| <b>Resolve checks</b>             | At 100 nerves a hero rolls: 75% Rattled (as now), 25% <b>Steeled</b> (a virtue: +accuracy, +dodge, and calms an ally each round). Steeled heroes are a reward for pushing the line                                                                                                                                                                                      | Nerve system                                                   |

| System                | What it adds                                                                                                                                                                                                                                                                            | Where it lives                                    |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Crew bonds</b>     | Heroes who finish expeditions together gain Bond (0-5). At 3+, they get a small shared perk (+5% damage when adjacent) and a chance for a supportive bark; heroes who watch a bonded friend die gain 20 nerves                                                                          | <code>Roster::bonds</code>                        |
| <b>The Drill Deck</b> | A Workshop tab where gold upgrades individual abilities (two levels each, +10% damage or effect per level, 60-180 gold). This is the mid-game gold sink the economy currently lacks                                                                                                     | Workshop panel                                    |
| <b>Elite enemies</b>  | Each base enemy has an elite variant from tier 3 (a name plate, +30% HP, one extra ability, a palette accent) so late fights aren't just bigger numbers                                                                                                                                 | <code>ScaleEnemyForTier</code>                    |
| <b>New enemies</b>    | Two per location: Cave: Barnacle Crab (guardian), Lantern Angler (blinds). Island: Fire Dancer (burn AoE), Idol Bearer (buffs). Weeds: Mantis Shrimp (armor break), Kelp Wraith (entangle, hidden). Atlantis: Drowned Oracle (foresight, heals), Star Spawn (madness, teleports a hero) | <code>enemyart.cpp</code> , <code>data.cpp</code> |

| System                      | What it adds                                                                                                                                                                                                                                                                                                                                                  | Where it lives              |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Deeper tiers</b>         | Two new locations unlocked by clearing all four Shallows at tier 6: <b>The Trench</b> (pressure debuff: –speed each round without a battery burn; boss: the Leviathan) and <b>The Hadal</b> (the final descent; the Abyssal Eye, a three-phase boss that reuses Cthulhu's staging tech). Each has its own ladder of 3 tiers at levels 5, 6, and 7 (a new max) | New <b>Location</b> entries |
| <b>Memorial and Sea Log</b> | The Library gains a memorial wall for dead crew (name, level, cause) and a Sea Log that records every boss kill, curio outcome, and first meeting with an enemy (with its dossier hint)                                                                                                                                                                       | Library panel, save file    |
| <b>Voyage events</b>        | Between expeditions, one event may fire in the salon: a passing salvager sells cheap relics, a storm damages the ship (one upgrade offline for a voyage), a crew member's habit flares, a card sharp challenges the dealer (a free Flats battle for gold)                                                                                                     | Salon on return             |

## Combat additions

- **Guard:** some skills (Hold the Line, Bulwark Mode) redirect hits from an ally for two rounds; the guarding figure steps in front in the stage.
- **Riposte:** the Captain's Steady Now and the Merman's Frenzy grant a counterattack against melee hits.
- **Crit reactions:** a hero crit removes 3 nerves from the whole party; an enemy crit adds 5 to everyone.
- **Enemy retreat:** Cowardly enemies can flee, denying loot but ending the fight.

### Balance targets after these systems land

- Sensible auto-player expedition win rates: 60 / 65 / 58 / 55 / 45% at cave levels 0 / 1 / 3 / 5 / 6, and 35% at level 7 in the Hadal.
- A full provisioning kit costs 15–20% of a typical expedition's payout.
- A hero who dies at level 4+ should feel like a 3-expedition setback, not a run-ending one.

### Definition of Done

- All systems in the table exist with data in `data.cpp`, are saved in `depth_save.txt`, and are covered by `--sim` (which now provisions, camps, and inspects curios sensibly).
- The Trench and the Hadal are playable with their bosses.
- The balance targets above are met or the deviations are logged.

### Flats: completing the visuals, card physics, and animation

Flats should feel like sitting at a real candlelit table in the salon: every card in the catalogue drawn, a dealer with hands and a face, bottles you can pick up, and cards that move with weight. The engine stays headless; all of this lives in `flats.cpp`, `flats_art.cpp`, and a new `flats_physics.cpp`.

### Art to complete

| Element            | Requirement                                                                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Every card         | All 32 catalogue cards, the 12 dealer-only cards, and the boss court drawn in the ink style with the concept-grid pipeline; each with a distinct silhouette and a 2-frame idle (a fin twitch, a blink) shown in the inspector |
| Card frame details | Tier shown by the driftwood frame's wear (clean, barnacled, bone-inlaid); editions add a foil sheen, a gilt edge, or a hex-mark burn                                                                                          |

| Element                | Requirement                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The table              | A round oak table with a worn felt inlay, the 4×4 board as brass-inlaid cells that glow when playable, candle wax pooling, the scales as a real brass balance with a hanging pan each side, the bell on a wooden block, the rules folder, and the pack as a leather roll of bottles                                                                                    |
| The room               | The salon behind the table, dim and out of focus (the same room as the hub, seen from the table); the great window's blue light on the left, candle on the right                                                                                                                                                                                                       |
| The dealer             | A full rigged figure across the table: torso, two hands, a face with eyes that track the player's cursor. Five dealers, each with a distinct look and a hand style (the Novice fumbles, the Tidewife's hands are wet and quick, the Wreck-Broker wears rings, the House has a mechanical hand, the Sovereign is a drowned crowned corpse whose court rises behind him) |
| Items                  | Each bottle drawn with its contents visible (a boulder, a goat, ink, a hook); uncorking shows the contents leaving                                                                                                                                                                                                                                                     |
| Map, nodes, and events | The run map as a sea chart under glass; every node event drawn as a small scene (the Maelstrom, the Boiling Vents, the Scrimshaw Artist's booth, the Splicers' tank, the stall's shelves, the campfire lantern)                                                                                                                                                        |
| Totems and charms      | Physical objects on the table's edge: totems as carved heads, charms hung on a cord above the scales                                                                                                                                                                                                                                                                   |

## Card physics ( `flats_physics.cpp` )

Every visible card has a `CardBody` : position, rotation, scale, velocity, and spring targets, simulated at 60 Hz independent of the engine's micro-steps. The engine's events set targets; the physics makes the motion.

| Moment                | Motion                                                                                                                                                               |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hand                  | Cards fan in an arc with spring spacing; hovering one lifts it 40 px, tilts it toward the cursor, and pushes its neighbors apart; the fan re-spaces when cards leave |
| Drag                  | The card follows the cursor with lag; its rotation comes from horizontal velocity (leaning into the move); a shadow grows with height                                |
| Drop                  | Snaps to the cell with an overshoot and a settle bounce; a small dust puff; illegal drops spring back to the hand                                                    |
| Sacrifice mode        | Candidate cards pulse and rise 10 px; a chosen card flips face-down and slides under the new card, leaving a blood drop on the felt                                  |
| Draw                  | The deck's top card flips through a 3D perspective quad (a textured <code>RL_QUADS</code> card with a back and a face) and arcs into the hand                        |
| Dealer plays          | The dealer's hand reaches out, places the card with a slight slap, and withdraws; queued cards advance with a slide                                                  |
| Strike                | The attacker lunges (existing), the target shakes and its <code>CardBody</code> recoils; a knocked-back card slides a row with a scrape                              |
| Death                 | The card cracks along the frame and dissolves into ink and bubbles; bones clatter into the bone pile                                                                 |
| Fledgling and Undying | Growth: the card swells, the frame reforms; Undying: a ghost copy rises and drifts to the hand                                                                       |
| Scales                | A real balance: pans dip with easing and overshoot, the needle swings and settles; at $\pm 7$ the whole scale trembles                                               |
| Bottles               | Pick up from the roll, tilt to aim, uncork; the contents fly in an arc to the target                                                                                 |

| Moment         | Motion                                                                                       |
|----------------|----------------------------------------------------------------------------------------------|
| Bell           | Depresses with the click and rings; the dealer reacts (nod, sneer, shrug) with a facial clip |
| Momentum boons | Coins spin onto the table; extra cards deal from the deck with a faster flip                 |

Dealer reactions

The dealer's face and hands respond to the game state: leaning in when ahead, drumming fingers while waiting, a flinch on a big hit, a slow clap on a loss, and a signature taunt animation per dealer at the start of each battle. Lines are shown as ink speech on the felt.

Definition of Done

- `--shots shots/Flats flats` renders every card, every dealer with three expressions, every item, every node scene, and both boss phases.
- A `--flats-ui-test` run shows every motion in the table above at least once and never leaves a card off its target.
- The dealer's eyes track the cursor and blink; the scales overshoot and settle; cards in hand re-space smoothly.

Flats: deeper systems and the long run map

Flats grows from a 9-layer sprint into a 15-layer run in three regions, with deck archetypes, side bets, dealer tells, and an ascension ladder, so it can hold a player for many runs. The existing rules, sigils, and catalogue remain the base.

The long run map

- **15 layers in three regions**, each ruled by dealers and ending in a boss table: **the Shallows** (layers 1–5, the Novice and the Tidewife, boss: the Tidewife's Reef Table), **the Reef** (6–10, the Wreck-Broker and the House, boss: the House), and **the Drowned Court** (11–15, elite tables, boss: the Atlantean Sovereign). Clearing the Court is clearing the run.
- **3–4 nodes per layer**, each connected to 1–3 nodes below, drawn as a sea chart with routes as currents. A route is a real commitment: once you enter a lane, the chart shows which nodes it can still reach.

- **Nodes show what they are, not what's inside.** Battles show the dealer; rewards show a type (card, charm, item, gold); event nodes show a glyph and a risk tag (safe, risky, dire).

**New node types**

| Node       | What happens                                                          | The decision                                                                                                    |
|------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Harbor     | The only place to cash out the pot, plus a small stall                | Cash-out nodes are rare (one per region); leaving with the pot means choosing the harbor lane over a richer one |
| Toll       | Pay 15% of the pot to pass, or fight a toll-keeper                    | Money now or risk now                                                                                           |
| Fog        | Contents hidden until entered                                         | Gamble on an unknown node for a lane with better rewards past it                                                |
| Elite lane | A visibly harder lane with a Tide Key at its end                      | The key opens a locked reward on the boss layer                                                                 |
| Undertow   | A current node that pushes you into a random adjacent node next layer | Speed and a bonus bone at the cost of control                                                                   |
| Wager      | Bet part of the pot on the next battle at 2:1                         | Double or nothing on a table you can see                                                                        |
| Salvager   | Trade one card from your deck for two random cards of its tier        | Deck thinning versus growth                                                                                     |

**Generation rules.** Every region has exactly one Harbor and at least one campfire and one stall; no two identical non-battle nodes are adjacent on a lane; every lane has at least two battles per region; exactly one elite lane per region; the boss layer is preceded by a breather. The generator rejects a map with a lane that has fewer than three rewards per region.

## Deeper systems

| System                   | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tribes matter</b>     | Coin, Cup, Blade, and Shell each get a passive when three of their cards are on your board: Coin pays 2 gold per slay, Cup heals your creatures 1 at turn end, Blade gives +1 strength to the front row, Shell gives +1 defense. Totems still add sigils by tribe                                                                                                                                                                                                                                                                                                                                                      |
| <b>Twelve new sigils</b> | Anchor (cannot be knocked back or shoved), Lure (the creature across must strike it), Ambush (strikes once when played), Cannibal (+1/+1 for each creature sacrificed to play it), Tide-Turner (+2 strength while the scales are against you), Mimic (copies the sigil of the creature to its left), Corrode (its target loses 1 defense permanently), Eclipse (the creature across cannot strike), Chorus (other Shell creatures +1 strength), Salvage (leaves an item when it dies), Hook (when played, pulls the queued enemy card across from it into your reserve), Wake (when it moves, adjacent enemies take 1) |
| <b>Deck archetypes</b>   | Rewards are weighted toward the tribes and costs already in the deck, so a Bone deck, a Blood-sacrifice deck, an Airborne deck, and a Big-fish deck each become viable by mid-run                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Sea Marks</b>         | A fourth reward class, earned only from boss tables: run-wide rules (Foresight: see the dealer's next two queued cards; High Tide: scales start +1 in your favor; Barnacled Deck: your deck's tier-1 cards gain +1 defense)                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Dealer tells</b>      | Each dealer has a hidden rule and a visible tell: the Tidewife strokes her hair before playing a Shell creature, the Wreck-Broker taps the anchor when his queue holds a heavy card, the House's mechanical hand clicks twice before a double draw. Learning tells is the skill ceiling                                                                                                                                                                                                                                                                                                                                |

| System                         | Detail                                                                                                                                                                                                                    |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Side bets</b>               | At the start of any battle, ante up to 25% of the pot on a condition (win in under 5 turns, win without losing a creature, slay 3) at 3:1                                                                                 |
| <b>Ascension: Depths</b>       | Clearing the run unlocks Depth 1 through 10: each adds a rule (dealers draw one extra card, the scales tip at 7, no minnow draws, harbors cost a card, and so on). The salon's card table shows the highest Depth cleared |
| <b>Deck limits and viewing</b> | Deck size 12–30; a deck viewer on the map screen; the salon lets the player save one deck as a starting favorite once it has cleared a run                                                                                |

## Balance targets

- The sensible bot clears the full 15-layer run about 5% of the time at Depth 0 (matching today's figure for the short run), wins region 1 about 70% of the time, and the auto-player's win rates per dealer stay in the same 45–97% band.
- Expected pot at a region-1 Harbor: about 120 gold; region 2: about 400; full clear: about 900 plus Marks. This keeps Flats the higher-variance income against the Periscope.

## Definition of Done

- `--flats-sim N` runs the long map with the new nodes and reports clear rate, region win rates, and average cash-out per region.
- `--flats-gen <seed>` prints a generated chart and asserts the generation rules.
- All twelve sigils, four tribe passives, Sea Marks, side bets, tells, and Depths 1–10 are implemented and shown in the rules folder.

## The Deep Arcade

The Deep Arcade is a cabinet in the salon that opens a menu of four multiplayer-only games. It is the only part of Depth with networking, and all four games share one networking layer, one lobby, and one visual frame.

## The cabinet

- Riveted into the foreground bulkhead: mahogany panels in tarnished brass, copper pipes running into the wall, a vent that puffs steam now and then.
- The screen is a thick bolted porthole glowing aqua, playing an attract loop of all four games in idle.
- The marquee is etched glass reading THE DEEP ARCADE, backlit by a tank of pulsing bioluminescent jellyfish.
- Controls are a ship's throttle lever for a stick and valve wheels for buttons; two pressure gauges show live data (players online, open lobbies).
- Clicking drops a brass token down a pneumatic tube with a thunk; the camera pushes into the porthole.

## The menu

- Inside the porthole: a rotating brass drum with four engraved reels, one per game. Scrolling clicks the drum; the selected reel lights and a preview loop, player count, match length, and one-line pitch show beside it.
- Three valve-wheel buttons: **Host** (opens a lobby and shows a 6-character join code), **Join** (type a code), **Browse** (open public lobbies and anything on the local network).
- **Lobby**: a crew roster board with 2–6 slots (per game), a display name, a faction or role pick, a ready lever, host settings, and a Launch handle that lights when everyone is ready. Chat is built in.
- **Settings gear**: display name, arcade audio, connection options.
- Winning an arcade match earns **arcade tokens** (10 per match, 25 per win) spent on cosmetics only: card backs, cabinet marquees, faction colors. The arcade never pays gold into the main game, so the economy stays as designed.

## Shared networking layer ( `arcade_net.cpp` , `arcade_lobby.cpp` )

| Piece   | Rule                                                                                                                   |
|---------|------------------------------------------------------------------------------------------------------------------------|
| Library | Valve's GameNetworkingSockets (C++): reliable and unreliable channels, encryption, NAT traversal. ENet is the fallback |

| Piece            | Rule                                                                                                                                                     |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model            | Host-authoritative for every game: the host's copy runs the real simulation; clients send commands and receive state. No deterministic lockstep anywhere |
| Channels         | Reliable ordered: commands, chat, lobby, diplomacy. Unreliable: state updates                                                                            |
| LAN              | The host broadcasts on the local network; Browse lists it                                                                                                |
| Internet, now    | A virtual-LAN app (ZeroTier or Tailscale) makes LAN mode work across the internet with no server                                                         |
| Internet, proper | A small lobby and relay server ( <code>relay/</code> , its own C++ program) on a cheap cloud host: join codes, hole punching, relay fallback             |
| Version check    | Same build and same data-file hash required to join                                                                                                      |
| Disconnects      | Pause for a reconnect (2 minutes); then an AI or a stand-in takes the seat                                                                               |
| Shared frame     | Every arcade game renders inside the same porthole bezel with the same lobby, pause, chat, and result screens                                            |
| Save             | Arcade profile (name, tokens, cosmetics, stats per game) stored in <code>depth_save.txt</code>                                                           |

Games on the drum

| Reel | Game       | Players             | Match length | One line                                                        |
|------|------------|---------------------|--------------|-----------------------------------------------------------------|
| 1    | Flats Duel | 2 (plus spectators) | 8-12 min     | Flats against a person: a best-of-three at the table            |
| 2    | The Trawl  | 1-4 co-op           | 15-25 min    | Fish the deep by night, fill the quota, don't wake what's below |

| Reel | Game    | Players | Match length | One line                                                          |
|------|---------|---------|--------------|-------------------------------------------------------------------|
| 3    | Scuttle | 2-4     | 5-10 min     | A fast, simple crab-racing card game anyone can learn in one hand |
| 4    | Fathoms | 2-6     | 20-30 min    | The island RTS (its own design document)                          |

## Definition of Done

- The cabinet, menu, lobby, and shared frame exist and can host and join Scuttle (the simplest game) over LAN.
- The relay server builds and runs; a join code connects two machines on different networks.
- Profiles and tokens save and load.

## Arcade game 1: Flats Duel

Flats Duel is Flats played against a person at the same table, best of three, using the headless `flats_board` engine with a symmetric board. The dealer's chair is taken by the opponent, drawn as their chosen dealer skin.

### Rules changes for two humans

- **Symmetric board.** Row 0 is the opponent's reserve, row 1 their front, row 2 your front, row 3 your reserve. There is no dealer queue; both players draw and play by the same upkeep rules (draw a card or a Minnow).
- **Turns alternate**, each with a 60-second timer (30 in Quick mode). Running out passes the turn.
- **Second-player balance.** The player who goes second starts with one extra bone and a Minnow in hand.
- **Scales tip at  $\pm 8$**  as usual. If a game reaches turn 20, the side the scales favor wins; at 0 it goes to sudden death (the next tip of any amount wins).
- **Banned or changed cards.** Massive, Tidal Pull, and the dealer-only cards are out. Undying returns at most twice per game. Spawn copies cannot Spawn. Sharp Quills caps at 3 damage per turn from one creature.
- **Items** are allowed, capped at 2 per player per game, chosen from the pack items.

- **Sideboard.** Between games in a best-of-three, each player may swap up to 3 cards from a 6-card sideboard.

## Modes

| Mode               | Decks                                                                                                                                        | Best for                             |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Draft<br>(default) | Before the match both players draft 20 cards from 8 shared packs of 4 (pick one, pass the rest), then add 2 Minnows and a Ballast Cask       | Fair matches between any two players |
| Constructed        | Each brings a 20-30 card deck built from cards they have unlocked in single-player; the host's game validates the list against the catalogue | Friends who both play Flats          |
| Quick              | Six preset decks (Bone, Blood, Airborne, Big Fish, Shell Wall, Swarm), no draft                                                              | A five-minute game                   |

## Presentation

- The opponent is a rigged figure across the table wearing a **dealer skin** chosen in the lobby (the five dealers plus skins bought with arcade tokens). Their hands place cards on their side; their face reacts to the game.
- Hidden information stays on the host: a client only ever receives its own hand, the board, both reserves' card backs, and the opponent's hand size.
- Six emotes (nod, sneer, shrug, tip hat, slow clap, gulp) on a cooldown, played by the opponent's figure.
- Spectators (up to 4) watch from the salon behind the table with both hands hidden.

## Definition of Done

- `--flats-duel-sim N` plays bot versus bot on the symmetric board; first-player win rate is 48-52% after the second-player bonus.
- Two clients complete a best-of-three over LAN with a draft, timers, sideboarding, and emotes.
- No client ever receives the opponent's hand contents (checked by a packet log in the test).

# Arcade game 2: The Trawl

The Trawl is a 1-4 player co-op fishing horror: a crew works a night trawler to fill the Owners' quota while the things under the boat try to take them. It takes Lethal Company's quota, deadline, scavenging, and dread, and How to Fish's cast-and-reel, gear, and grounds. It reuses the parkour side's pixel-art pipeline and beast sprites, so it is built mostly from existing parts.

## The loop

- 1. **Dock.** The crew meets on the trawler at the Owners' pier. The quota and the nights left are chalked on a board. Buy gear at the chandler.
- 2. **Sail.** The helm picks a fishing ground; the trip is a short top-down sail through fog.
- 3. **The night** (about 5 minutes). Work the deck stations, land fish and salvage, survive.
- 4. **Return and sell.** Fish are weighed, salvage valued; money goes to the quota first, the rest to gear.
- 5. Repeat for three nights, then the quota is due. Meet it and the quota rises; fail and the run ends.

A session is three nights plus shopping, about 20-25 minutes.

## The deck (top-down, pixel art)

The trawler is a small top-down deck at night; only lantern light is visible, and the sea beyond the rail is black. Stations:

| Station       | What a player does                                                                                                       |
|---------------|--------------------------------------------------------------------------------------------------------------------------|
| Helm          | Steers; holding position matters for nets and dives                                                                      |
| Lines (2)     | Cast and reel: a tension bar that must be kept in the green while the fish runs; heavier fish mean longer, wilder fights |
| The net winch | Drops and hauls the trawl net; big catch, but the net drags and can snag on wrecks or on something alive                 |
| Lantern mast  | Raises or lowers the light: more light means more fish and more attention from below                                     |

| Station        | What a player does                                                                                                                                                               |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sonar          | Pings show fish schools and large shapes; too many pings wake the large shapes                                                                                                   |
| Harpoon cannon | Bought gear: one shot per reload against whatever surfaces                                                                                                                       |
| Diving bell    | Bought gear: one crewman dives a side-view mini-level (parkour movement, no jumping hazards) to a wreck for high-value salvage while the deck crew keeps the boat over the wreck |
| The bell       | Anyone can ring it: the crew's alarm and the way a dead player's ghost can still warn the living                                                                                 |

## What's below

Threats grow with the night's minutes and the lantern level. Each has a tell, a counter, and a cost.

| Threat          | Tell                                 | What it does                                                 | Counter                                     |
|-----------------|--------------------------------------|--------------------------------------------------------------|---------------------------------------------|
| Ghost Worm      | Sonar shows a long shape circling    | Bites lines, snags the net                                   | Cut the line, lower the lantern             |
| Lantern Angler  | A second light in the dark           | Lures a crewman to the rail and pulls them over              | Don't walk toward lights you didn't hang    |
| Sirens          | Singing, the helm drifts             | Turn the wheel toward rocks; charm a player for 5 s          | Plug ears (gear), ring the bell             |
| The Great White | A fin in the lantern's edge          | Rams the hull, tips crew into the water                      | Harpoon cannon, full speed away             |
| Drowned sailors | Knocking on the hull                 | Climb aboard; slow, relentless                               | Hold them off with gaffs, keep the deck lit |
| Kraken          | The sea goes still; sonar whites out | Tentacles grab crew and gear off the deck; can take the boat | Cut the net, kill the lantern, run          |

| Threat     | Tell                   | What it does                      | Counter                              |
|------------|------------------------|-----------------------------------|--------------------------------------|
| Ghost ship | A bell answering yours | Boards with a crew of the drowned | Escape or fight; loot if it's beaten |

A player taken by the sea is dead for the night; their ghost can walk the deck, ring the bell, and read the sonar. They return at the dock if the boat comes home.

Fishing grounds and gear

| Ground          | Fish value | Salvage            | Danger                         |
|-----------------|------------|--------------------|--------------------------------|
| The Shallows    | Low        | Little             | Ghost Worms                    |
| The Kelp        | Medium     | Nets snag          | Sirens, sharks                 |
| The Wrecks      | Medium     | High (diving bell) | Drowned sailors                |
| The Trench      | High       | Rare relics        | Anglers, the Kraken            |
| Atlantis Waters | Very high  | Very high          | Everything, and the ghost ship |

Gear (bought with money past the quota): stronger lines, a bigger net, a brighter or hooded lantern, the harpoon cannon, the diving bell, hull plates, ear plugs, gaffs, and a second sonar. The Owners raise the quota by about 40% after each successful deadline, so the crew has to move to riskier grounds.

Definition of Done

- `--trawl-sim N` runs a bot crew through three nights per ground and reports quota-met rates: about 80% on the Shallows and 30% on Atlantis Waters with starting gear.
- Four players over LAN complete a session with a dive, a Kraken attack, and a death-and-ghost.
- Every threat has a distinct pixel sprite, sound tell, and tested counter.

Arcade game 3: Scuttle

Scuttle is a 2–4 player crab-racing card game that takes one hand to learn and about seven minutes to play. Each player's crab scuttles along an 8-space tide pool; you play one card a turn to move crabs, mess with the others, and bet on who reaches the tide line. It

reuses Flats' card art, table, and physics with a smaller ruleset.

Setup

- Each player has a crab (a colored shell) on space 0 of a shared 8-space tide pool drawn on the felt.
- A shared deck of 60 cards is shuffled; each player draws 5.
- A round is over when any crab reaches space 8. A match is first to 2 round wins (3 for a longer game).

A turn

1. Play one card from your hand and resolve it.
2. Optionally place one bet token (see Bets) if you have any left.
3. Draw back to 5.

The deck

| Card              | Copies     | Effect                                                                                       |
|-------------------|------------|----------------------------------------------------------------------------------------------|
| Scuttle 1 / 2 / 3 | 10 / 8 / 4 | Your crab moves forward that many spaces                                                     |
| Sidestep          | 6          | Swap your crab with any crab one space ahead                                                 |
| Wave              | 5          | Every crab except yours moves back 1                                                         |
| Pinch             | 5          | Target crab skips its owner's next move card                                                 |
| Shell             | 5          | Play in response to any card aimed at you to cancel it                                       |
| Current           | 4          | Every crab moves forward 2; the crab in last place moves 3                                   |
| Gull              | 4          | The crab in the lead moves back 2                                                            |
| Rock              | 3          | Place a rock on a space; the next crab to land there stops for a turn, then the rock is gone |
| Molt              | 3          | Discard your hand and draw 5                                                                 |
| Tide Rush         | 3          | Move your crab 1 for each other crab ahead of it                                             |

## Bets

Each player has 3 bet tokens per round. A token is placed face down on any crab (yours or another's) and pays 3 points if that crab wins the round, 1 if it comes second, and -1 otherwise. Winning the round with your own crab also pays 2 points. The match winner is decided by round wins; points break ties and feed the arcade-token payout.

## Presentation

- The tide pool is drawn on the Flats felt with pixel-crab tokens that scuttle sideways between spaces, kick sand, and flip over when Pinched.
- Cards use the Flats frame with a shell-and-sand back. Bet tokens are pearls.
- A 20-second turn timer keeps it brisk; an empty seat is played by a simple bot.

## Definition of Done

- `--scuttle-sim N` shows no seat position wins more than 30% in 4-player games and the average match lasts 6–9 minutes of turns at the timer's pace.
- Four players over LAN finish a match with bets and a bot filling a dropped seat.

## Arcade game 4: Fathoms

Fathoms is the 2–6 player steampunk island RTS. Its complete design, with every unit stat, faction mechanic, neutral power, victory rule, balance result, and its own build roadmap, lives in `Fathoms — RTS Design Document`. This section only places it in the arcade.

**In one paragraph.** Players start on home islands, gather Food, Brass, Coal, and Ichor, claim islands with Lighthouses, and advance through the Sail, Steam, and Leviathan eras. Six factions drawn from Depth's locations each play differently: the Nautilus Crew (tech cards and submarines), the Islanders (ally tribal towns), the Crustacean Brood (molting, seafloor walking), the Merfolk (kelp spread, swimming armies), the Atlantean Lost Ones (Devotion, conversion, a Cthulhu avatar), and the Clockwork Foundry (a coal-powered army that never breaks). Pirates can be traded with or hired against rivals, tribes raid or ally, volcanoes erupt over the richest Ichor, and a Kraken wakes at minute 20. Matches end by conquest, relics, the volcano altar, or score at a 30-minute cap.

## How it fits the arcade

- Uses the shared networking layer, lobby, and frame; adds the diplomacy panel and a headless-server build option described in its document.

- Its art is Darkest Dungeon ink at a smaller scale: units are simplified rig figures with the same outline and hatching rules; the sea, islands, and weather use the parallax prop system.
- Its balance data file and simulator are the model for the other arcade games' balance files.
- Player-facing tie-ins: faction skins and unit colors are bought with arcade tokens; boss kills in Fathoms (the Sun God, Blackbeard, the Kraken) are recorded in the Sea Log.

**Order of work.** Build it last of the four games, after the shared layer has shipped with Scuttle and Flats Duel and been tested over the internet, because it stresses the networking the most (six players, hundreds of units, fog filtering).

## Sound design

Everything stays synthesized in code, extending `sound.cpp` and `sound_parkour.cpp`. The rule for every scene: a bed of ambience that never stops, music that changes with the state of play, an effect for every action, and one signature sound per character, creature, and station.

### Architecture ( `sound.cpp` )

- **Voices.** Subtractive (oscillators plus filters, for tones and drones), FM (bells, brass, glass), noise-shaped (water, wind, steam, hisses), physical-model plucks and taps (strings, wood, card edges), and formant (voices and songs). Add a generated convolution reverb: an impulse per room type (salon, cave, kelp, hall, open sea) built from filtered noise decays.
- **Buses.** Music, Ambience, Effects, UI, Voice. Effects duck music by 3 dB during combat impacts; Voice ducks everything by 4 dB. Separate volume sliders for Music and Effects stay; add Ambience.
- **Cue registry.** Every sound is a named cue in `data.cpp` (pitch range, variations, bus, priority, max simultaneous). `--audio-test` plays every cue in order and fails on any that is silent or clips.
- **Variation.** Every effect has 3–6 variants by random pitch ( $\pm 6\%$ ), filter, and length so nothing repeats exactly.
- **Priority.** At most 24 effect voices; low-priority cues (footsteps, drips) are dropped first.

## Music

| Where                    | Style                                                                       | How it changes                                                                                                                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Salon                    | A slow Verne-era waltz on harmonium and music box over the engine's low hum | Adds a station motif when a panel opens (organ chord for the Sick Bay, a sonar arpeggio in the Radar Room, a shuffle rhythm at the card table); mutes to the organ alone when a crew member has died this voyage |
| Expedition walking       | Sparse: a drone per location and a slow pulse                               | Pulse quickens as light drops; a low tone rises the closer the party is to the boss room                                                                                                                         |
| Expedition combat        | Location palette, tighter rhythm                                            | Layers: base rhythm, a melodic layer while the party is winning, a dissonant layer when any hero is below 30% HP or above 80 nerves, a heartbeat at Death's Door; a stinger on crits and kills                   |
| The Cave                 | Glass harmonica and dripping percussion                                     | Mould glow pulses on the beat                                                                                                                                                                                    |
| The Island               | Log drums, conch, chant formants                                            | The chant rises when the Shaman heals                                                                                                                                                                            |
| The Weeds                | Bowed strings, whale calls, kelp creaks                                     | Whale calls answer the Siren's abilities                                                                                                                                                                         |
| Atlantis                 | Choir formants, a sub-bass organ, a ticking that isn't a clock              | The organ chord changes when Cthulhu's phase changes                                                                                                                                                             |
| The Trench and the Hadal | Pressure drone, metallic groans, near silence                               | Hearing the ship's hull creak here is the point                                                                                                                                                                  |
| Bosses                   | A theme per boss built from its location palette, with a phase-2 variation  |                                                                                                                                                                                                                  |

| Where       | Style                                                                   | How it changes                                                                                               |
|-------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Flats       | Muted harmonium and a music box, candle crackle in the mix              | Speeds up when the scales pass $\pm 5$ ; a drowned-choir layer for the Sovereign; silence when Selenis rises |
| Arcade menu | A chip-tune-styled synth loop with a music-box lead, CRT hum underneath |                                                                                                              |
| The Trawl   | A drone and a slow shanty fragment that stops whenever a threat is near | The music stopping is the first tell                                                                         |
| Scuttle     | Bright plucks, a ukulele-like synth and shakers                         |                                                                                                              |
| Fathoms     | A theme per faction and per era, blending as the game advances          | A tremor rumble under the music before an eruption; a Kraken motif at minute 20                              |

## Ambience

| Place       | Bed                                                                                                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Salon       | Engine hum, hull creaks with the roll, a wall clock, water against the window, the cat's purr near the cat, the organ's breath in the Sick Bay corner, distant footsteps of the hands |
| Cave        | Drips with reverb, distant rockfall, mould hiss, water lapping                                                                                                                        |
| Island      | Surf, insects, drums far off, palm rustle, a dog barking                                                                                                                              |
| Weeds       | Kelp creak, bubbles, whale song, fish flutter                                                                                                                                         |
| Atlantis    | A low choir, chains, stone grinding, the eye's hum                                                                                                                                    |
| Flats table | Candle, the felt, the dealer's breathing, the salon muffled behind                                                                                                                    |

| Place      | Bed                                                                                     |
|------------|-----------------------------------------------------------------------------------------|
| Trawl deck | Waves, wind, rigging, the engine, the winch, knocking on the hull when the drowned come |
| Fathoms    | Sea, gulls near islands, drums near tribes, pirate cove revelry, volcano rumble         |

## Effects by area

- **Hub.** Station hovers: brass plaque slide, the wheel's creak, the sonar sweep, the organ's two notes, the candle's flare, the arcade's jellyfish hum and gauge tick, the hatch wheel turning and the ladder's rungs. Panels: a whoosh and a brass latch. Embark: the engine spooling and the hull tilting.
- **Expeditions.** Footsteps per surface (stone, sand, kelp, marble), breathing in the diver's helmet, the flashlight's click and hum, the sonar scope's ping and static. Combat: a hit family per attack (slash, thrust, blunt, shot, throw, cast, song), a distinct impact per material struck (flesh, shell, metal, stone), a crit crack, a dodge whoosh, a stun's gear spin, bleed drips, poison bubbling, the heal chime, the nerve ripple, the Death's Door heartbeat, and each enemy's voice, pain, and death. Bosses: a roar per boss and a phase-change stinger.
- **Flats.** Card slide, flip, snap onto felt, shuffle, the fan re-spacing, sacrifice cut, bones clatter, the scales' creak and needle tick, the bell, bottle uncork and contents, card death crack and bubbles, Fledgling growth, Undying's ghost hum, and a formant mumble per dealer (each with its own pitch and rhythm) for their lines.
- **Arcade.** The token's thunk down the tube, the drum's clicks, valve wheel turns, the porthole's CRT hum, lobby lever pulls, the Launch handle. Trawl: the reel's ratchet under tension, the line snapping, the winch, sonar pings, the bell, the siren's song, the angler's second-light hum, the Kraken's silence then wood splintering. Scuttle: crab scuttles, sand kicks, pearl tokens. Fathoms: unit acknowledgments per faction (a whistle for the Crew, a drum for the Islanders, clicks for the Brood, a song for the Merfolk, a chant for the Atlanteans, a steam hiss for the Foundry), cannon fire, torpedo whine, era fanfares, the tremor and eruption, map pings.
- **UI.** Hover ticks, confirm and cancel plates, a low error thud, the level-up brass fanfare, the results panel's chalk.

## Accessibility

- Every sound tell (a dealer’s tell, a Trawl threat, an eruption tremor) also has a visual cue.
- A subtitles toggle shows barks and dealer lines as text.

## Definition of Done

- The cue registry lists every cue in this section; `--audio-test` passes.
- Every scene has an ambience bed and a music state machine with at least two states.
- Every action a player can take makes a sound, and every creature has a voice, pain, and death cue.

## Build order and testing

Do the shared visual foundation first, because the hub, expeditions, and Fathoms all depend on it; then the single-player systems; then the arcade from simplest game to hardest. Each stage ends at its section’s Definition of Done.

| Stage | Work                                                                                                                   | Section                  | Gate before moving on                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------|
| 1     | Shared rig, clips, chains, three-light rig, palettes, <code>--silhouette</code>                                        | Shared visual direction  | One hero and one enemy fully rebuilt and approved side by side with a Darkest Dungeon screenshot |
| 2     | The salon rebuilt in depth with all stations, the arcade cabinet (menu not yet wired), the study hatch and placeholder | The hub                  | Hub Definition of Done                                                                           |
| 3     | Sound architecture: buses, cue registry, reverb, salon ambience and music states                                       | Sound design             | <code>--audio-test</code> passes for the salon                                                   |
| 4     | Sonar chart, room and corridor generation, objectives                                                                  | Expeditions: sonar chart | <code>--gen-chart</code> and <code>--sim</code> pass                                             |

| Stage | Work                                                                                                                                                | Section                      | Gate before moving on                                        |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------|
| 5     | Crew, enemy, boss, and background overhaul on the rig; combat HUD; effects language                                                                 | Expeditions: visual overhaul | Sheets for every figure in <code>--shots</code>              |
| 6     | <code>EnemyBrain</code> , personalities, boss scripts                                                                                               | Expeditions: enemy AI        | <code>--brain-test</code> numbers logged                     |
| 7     | Provisioning, curios, camping, habits, ailments, resolve, bonds, Drill Deck, elites, new enemies, the Trench and the Hadal, memorial, voyage events | Expeditions: new content     | Balance targets met                                          |
| 8     | Expedition sound: location music, combat layers, every cue                                                                                          | Sound design                 | <code>--audio-test</code> passes for expeditions             |
| 9     | Flats art completion, card physics, dealer rigs                                                                                                     | Flats visuals                | <code>--flats-ui-test</code> shows every motion              |
| 10    | Long run map, new nodes, sigils, tribes, Marks, tells, side bets, Depths                                                                            | Flats systems                | <code>--flats-sim</code> and <code>--flats-gen</code> pass   |
| 11    | Arcade menu, lobby, shared networking layer, relay server, profiles; Scuttle as the first game                                                      | Deep Arcade, Scuttle         | Two machines on different networks play Scuttle by join code |
| 12    | Flats Duel                                                                                                                                          | Flats Duel                   | Best-of-three over the internet with draft                   |
| 13    | The Trawl                                                                                                                                           | The Trawl                    | Four-player session over the internet                        |
| 14    | Fathoms, following its own document's eight stages                                                                                                  | Fathoms                      | Six-player match over the internet                           |
| 15    | Arcade sound and remaining cues; final mix pass                                                                                                     | Sound design                 | Full <code>--audio-test</code> passes                        |

## Testing tools to add to `depth.exe`

| Tool                                                | Checks                                                                        |
|-----------------------------------------------------|-------------------------------------------------------------------------------|
| <code>--silhouette [filter]</code>                  | Renders figures solid black for silhouette review                             |
| <code>--gen-chart &lt;tier&gt; &lt;seed&gt;</code>  | Prints an expedition chart and asserts the generation rules                   |
| <code>--brain-test &lt;tier&gt; &lt;runs&gt;</code> | Hero win rate against <code>EnemyBrain</code> per tier                        |
| <code>--flats-gen &lt;seed&gt;</code>               | Prints a Flats run chart and asserts its rules                                |
| <code>--flats-duel-sim N</code>                     | Bot-versus-bot on the symmetric board                                         |
| <code>--trawl-sim N</code>                          | Bot crew quota rates per ground                                               |
| <code>--scuttle-sim N</code>                        | Seat fairness and match length                                                |
| <code>--net-loop &lt;game&gt;</code>                | Runs a host and two clients in one process over loopback for a scripted match |
| <code>--audio-test</code> (extended)                | Plays every registered cue                                                    |

## Working rules for Claude Code

- One stage per task; read the section, the relevant existing design doc, and `CLAUDE.md` before touching code.
- Every stage adds its self-test before its feature is called done, and logs balance numbers in a changelog.
- Numbers live in data files; art stays procedural; the parkour side is not touched.
- When a section’s design and the code disagree, the code’s existing self-tests win until the design is updated here.

## Open questions for the designer

- ☐ Should the Study hatch be visible from the start or unlock after the first expedition?
- ☐ Should arcade tokens ever unlock anything in single-player (a card back for Flats, a hero skin), or stay arcade-only as written?
- ☐ Is level 7 (the Hadal) the final crew level, or should the cap rise again later?